Applied Software Project Management

Project Schedules

What is a project schedule?

- Core of a project plan
- The *project schedule* is a calendar that links the tasks to be done with the resources that will do them.
  - Before a project schedule can be created
    - The project manager must have a work breakdown structure (WBS) and estimates.
    - An effort estimate of each task
    - Resources list with availability of each resource

Project Schedule cont.

- A PM time is better spent on working with team to create WBS and estimates than trying to build a schedule without them
- Schedule itself is an estimate
- Many tools and books on them to teach how to use these tools however for PM it is important to Understand concepts behind:
  - WBS and Dependencies
  - Resource Allocation and critical paths
  - Gantt Chart and Earned value analysis

Resource Allocation

- First step – Identify resources required
- Resources could be – People, Computer resources, locations, services and equipment temporarily acquired for the project
- Resources must be allocated to each task
- People capability -> task assignment
- Single person single task, the other task must wait if require same capability
- Parallel – like programming task divided b/w 3 programmers
- Effort (persons / hour) & duration (days, weeks etc.)

Scheduling concepts: Effort vs. Duration

- Effort represents the work required to perform a task.
  - Effort is measured in person-hours (or person-days, person-weeks, etc.)
  - It represents the total number of hours that each person spent working on the task.
- Duration is amount of time that elapses between the time the task is started and the time it is completed.
  - Duration is measured in hours (or days, weeks, etc.)
  - It does not take into account the number of people performing the task

Resource Allocation cont.

- One resource allocation to two tasks by percentage of resource time to each task
- If more than one person is allocated to a task PM must take overhead into account
  - "Overhead effort that does not go to the core activities of the task but is still required in order for the people to perform it"
  - For example, if 1 person perform a task in 12 days, 2 will do it in 7 days. 1 day effort in integration
  - Senior team members are scarce, juniors take more time, best to combine both in a task if possible

Resource Allocation cont.

- Two useful and well-known principles while working on projects.
- Parkinson's Law states, "Work expands so as to fill the time available for its completion."
- Fred Brooks pointed out in his 1975 book The Mythical Man-Month,

  "Nine women cannot have a baby in one month"
- In other words, some tasks can be done only by one person, no matter how critical that task is

Scheduling concepts: Slack and Overhead

- Overhead already discussed
- Slack is the amount of time which any of the tasks can be delayed without causing the due date of the final task in the sequence to be delayed as well.
  - A tight schedule has very little slack; a delay in any task will cause a delay in the due date
  - Parkinson's Law: "Work expands so as to fill the time available for its completion."

Building the project schedule: Steps

- Allocate resources
  - For each task in the WBS, one or more resources must be assigned
  - Choose person or people for each task based on qualifications, familiarity and availability
  - Take overhead into account when calculating the duration of each task

Building the project schedule

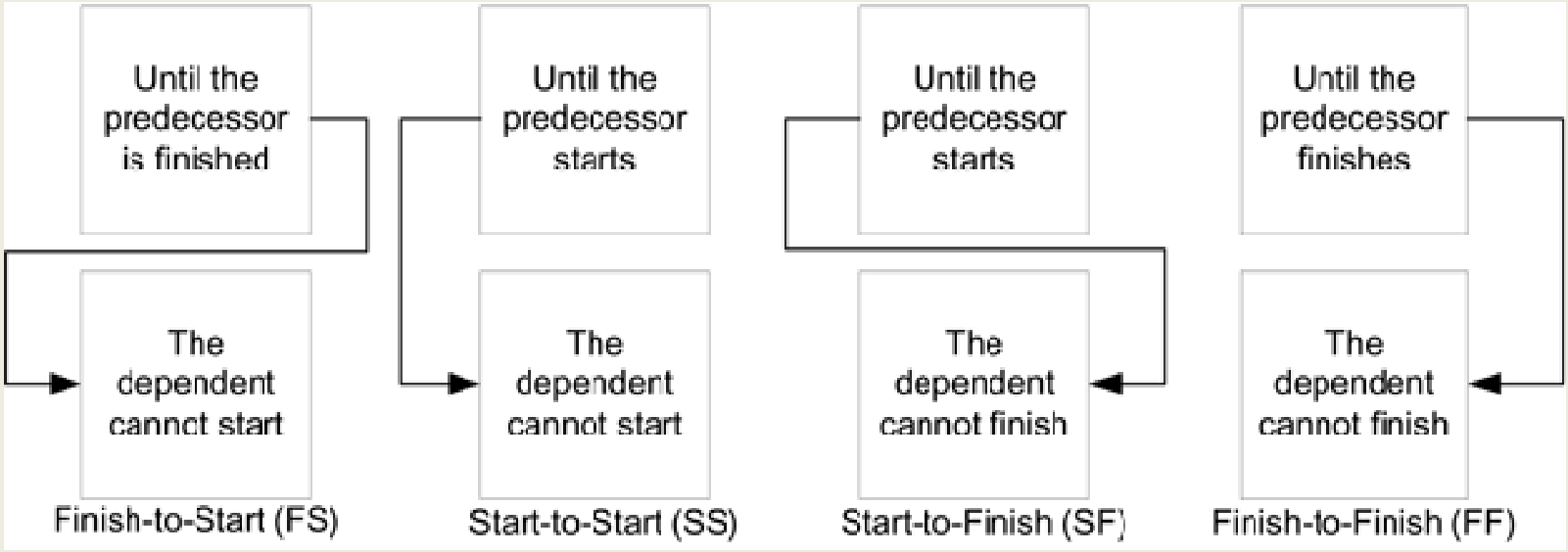
- Identify dependencies
  - A task has a dependency if it involves an activity, resource or work product which is subsequently required by another task
  - Tasks may have dependencies because they require the same resource
  - Dependencies forms:
    - A test plan can't be executed until a build of the software is delivered
    - Code might depend on classes or modules built in earlier stages
    - A user interface can't be built until the design is reviewed.

Building the project schedule: Steps

- Identify Dependencies cont.
  - It is PM responsibility to work with everyone on the engineering team to identify these dependencies.
  - The PM should start by taking the WBS and adding dependency information to it:
  - Each task in the WBS is given a number, and the number of any task that it is dependent on should be listed next to it as a predecessor

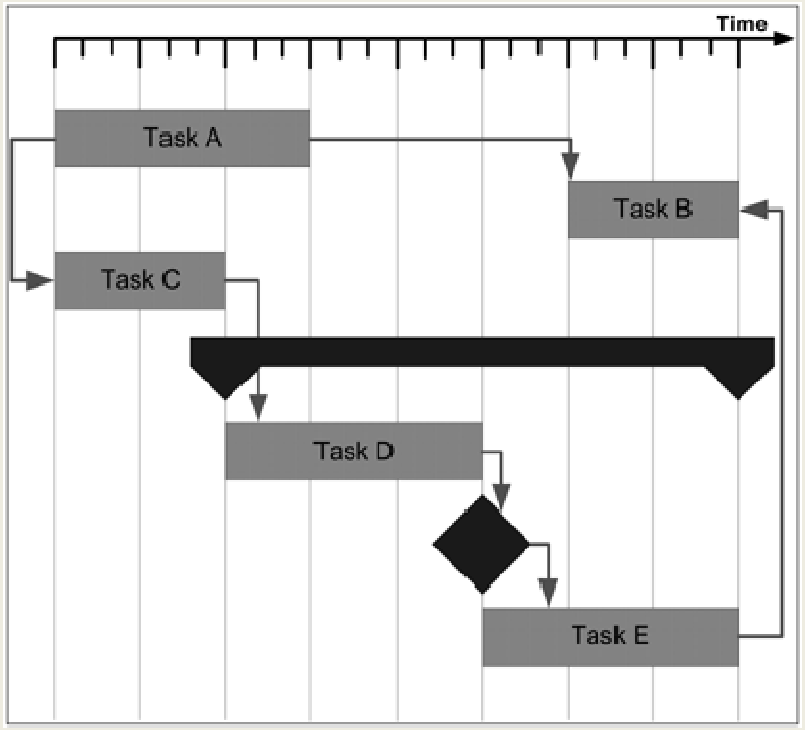
Building the project schedule

- Identify dependencies (cont.)
  - Every dependency has a *predecessor*, or a task that must be begun, in progress, or completed, for another task to begin
  - Identify the type of predecessor for each dependency



Building the project schedule

- Create the schedule
  - Most project schedules are represented using a Gantt chart
  - The Gantt chart shows tasks, dependencies and milestones using different shapes

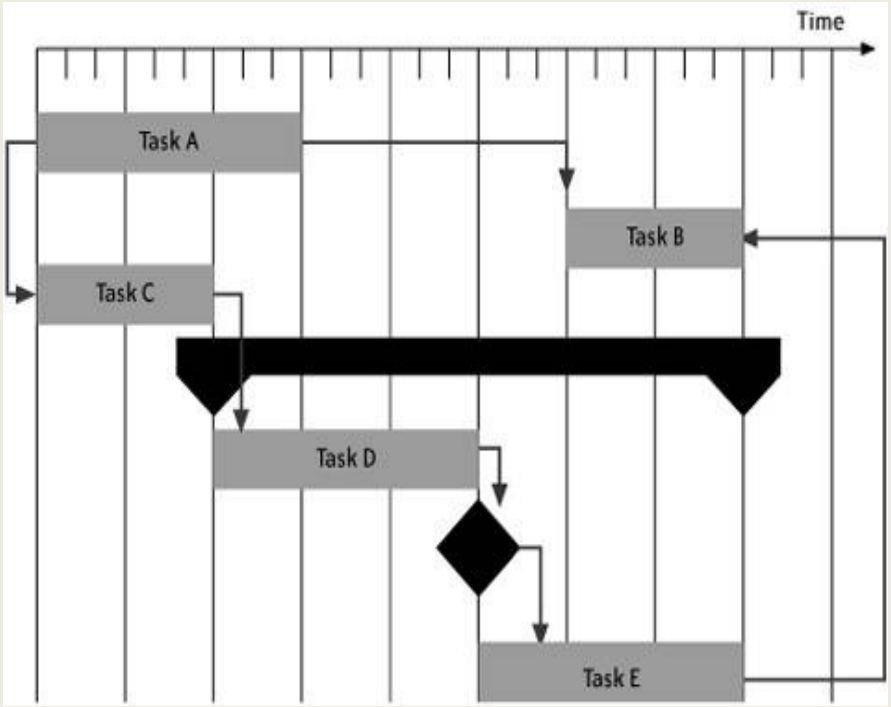


Building the project schedule:

- Create Schedule cont. : Gantt Charts
  - Each task is represented by a bar, and the dependencies between tasks are represented by arrows
  - Each arrow either points to the start or the end of the task, depending on the type of predecessor
  - The black diamond between tasks D and E is a milestone, or a task with no duration
  - Milestones are used to show important events in the schedule.
  - The black bar above tasks D and E is a summary task, which shows that these tasks are two subtasks of the same parent task

Building the project schedule: Gantt Charts

- Task A is a Finish-to-Start (FS) predecessor of Task B
- Task A is a Start-to-Start (SS) predecessor of Task C
- Task C is a Finish-to-Start (FS) predecessor of Task D. Task D is a Finish-to-Start (FS) predecessor of a milestone, which in turn is a Finish-to-Start predecessor of Task E.



Building the project schedule: Gantt Charts

- Task E is a Finish-to-Finish (FF) predecessor of Task B. Note that there is a delay before Task B, it does not start until its planned completion time will match up with Task E.
- This allows the resources required for Task B to be allocated to another task in the meantime.
- For example, a test plan can be started as soon as the requirements are complete, but it cannot be completed until after the design is done.
- So the test plan task would have the requirements task as a Finish-to-Start predecessor and the design task as a Finish-to-Finish predecessor.

Building the project schedule

- Reconcile the schedule with the organization's needs
  - Once resources are allocated to each task, a final date can be calculated
  - If this date is unacceptable, the project plan must change
  - Either additional resources must be allocated to the project or the scope must be cut down
  - Brooks' Law: "Nine women cannot have a baby in one month."
    - In other words, some tasks can only be done by one person, no matter how critical they are.

Building the project schedule

- Reconcile the schedule with the organization's needs: Ways to reconcile are:
  - If due date does not fit with the needs of organization or stakeholders, PM should first look to reallocate resources more efficiently
  - Look for the large gaps in schedule
  - Secondly to add or rearrange resource if available
  - Thirdly technical solutions like revising original estimation. i.e. Instead of not complementary user interface build a command line interface, or buying off the shelf component instead of building

Building the project schedule

- Add review meetings to the schedule
  - If there is a project plan, it should be consulted again and again
  - Progress reviews are meetings held regularly to check the progress of a project versus it's scheduled progress.
  - Milestone reviews are meetings which the project manager schedules in advance to coincide with project events.
    - The most common way for project managers to handle milestone reviews is to schedule them to occur after the last task in a project phase (such as the end of design or programming).

Building the project schedule

- Add review meetings to the schedule cont.
  - If a task is going to be late the PM must find a way to deal with it
  - Delayed tasks may cause other tasks to be delayed
  - Therefore, stakeholders must be present and can help to resolve major problems
  - Serious problems if cannot be dealt within schedule than:
    - Team have to adjust schedule to put in over time or
    - Team should go back to vision and scope document and scale back scope of the project

Building the project schedule

- Add review meetings to the schedule cont.
  - The schedule should always be updated by the PM but should be agreed upon at status meetings.
  - After every status meeting, the schedule should be updated with any changes that were agreed upon in the meeting.
  - If a team member discovers a problem between status meetings that could cause a delay, the PM should be notified immediately.
  - If the problem is serious enough not worth waiting, a new meeting should be called, and the team should talk about its impact to update schedule

Building the project schedule

- Optimize the schedule
  - PM have option to rearrange the schedule from the flexibility by which the tasks may be performed
  - In many schedules, there is some slack in tasks sequences. However a tight schedule has a very little slack. A delay in a task cause delay in due date
  - Even if there is a slack, all of the resources could be allocated to another task in the slack period.
  - PM's fall into the trap of using slack in the schedule as a way to mitigate risk

- Optimize the schedule
  - PM think second task will have some protection, in case the first task is late.
  - PM’s who try to do this find that Parkinson's Law kicks in the work in the first task expands to fill up the slack.
  - Instead of using slack to mitigate risk, the definition of the task should be expanded to include an activity that may mitigate the risk

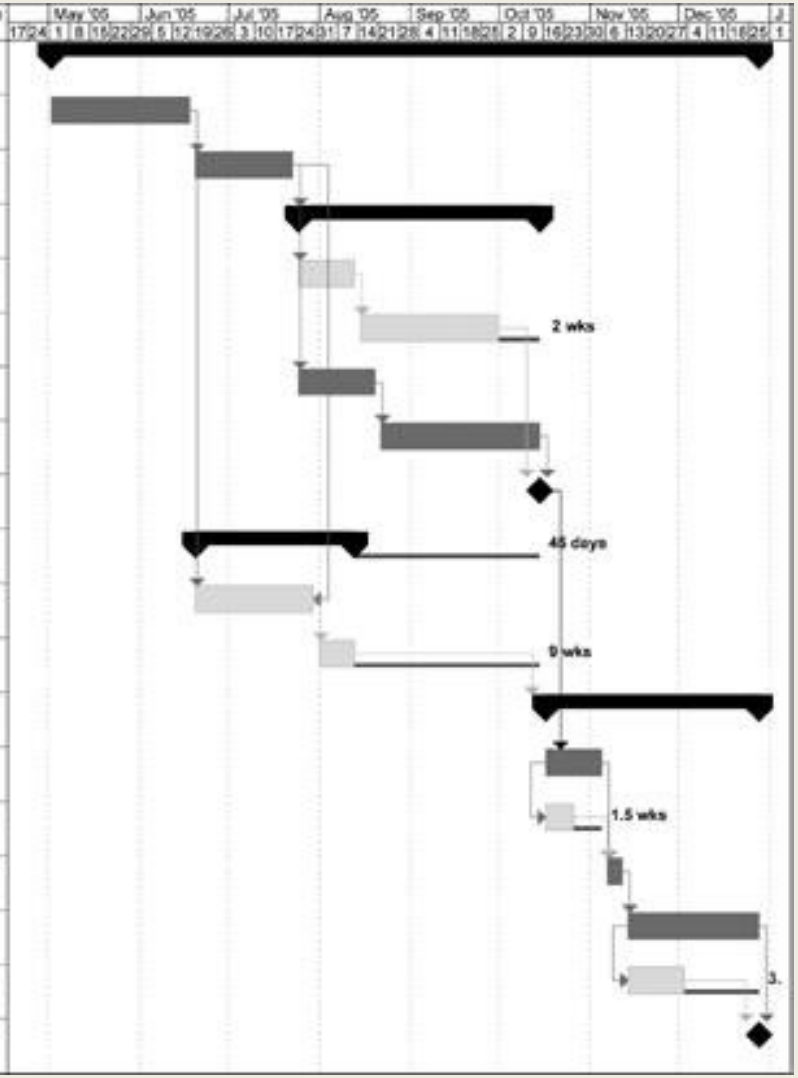
Building the project schedule

- Optimize the schedule
  - The critical path is the sequence of tasks that represent the minimum time required to complete the project.
    - If a task is only on the critical path when delaying that task will delay the project.
    - Allocating resources to tasks on the critical path will reduce the project schedule; allocating them to other tasks (not on critical path) will have less effect.

Building the project schedule

- Example of project schedule showing critical path

ID	Task Name	Duration	Predeces
1	**Software Project**	**172.5 days**	
2	Requirements	7 wks	
3	Design	5 wks	2
4	**Programming**	**60 days**	**3**
5	Unit Tests for Feature A	3 wks	3
6	Program Feature A	7 wks	5
7	Unit Tests for Feature B	4 wks	3
8	Program Feature B	8 wks	7
9	Feature-Complete Build	0 days	6,8
10	**Test Preparation**	**40 days**	
11	Build Test Plans	6 wks	2,3FF
12	Review, Correct Test Plans	2 wks	11
13	**Test Execution**	**52.5 days**	**12**
14	Execute Test Plan A	3 wks	9
15	Execute Test Plan B	1.5 wks	14SS
16	Fix Defects	1 wk	14,15
17	Regress Test Plan A	6.5 wks	16
18	Regress Test Plan B	3 wks	17SS
19	Deliver Beta Build	0 days	17,18



Building the project schedule

- In the above example, the test preparation tasks are not on the critical path.
- This means that if there is a delay in building or reviewing the test plans, then the project due date will not change unless that delay is long enough to put those tasks back on the critical path.
- In this case, that would require nine weeks (Thin bars to the right of test preparation tasks).
- The "1.5wks" label next to the "Execute Test Plan B" task shows that task would have to be delayed by 1.5 weeks to impact the due date.)

Building the project schedule

- A resource is over-allocated if more than 100% allocated to multiple tasks simultaneously
  - If any resource is over-allocated, it means that there is a dependency between two tasks which was not discovered.
  - When this happens, the schedule is guaranteed to be inaccurate. Find and fix over-allocated resources.

Building the project schedule

- Don't Abuse Buffers
  - A *buffer* is a task added to the schedule with no specific purpose except to account for unexpected delays.
  - This practice involves either adding extra tasks or padding existing tasks at strategic points in the schedule where overruns are “expected”.
  - Buffers can be useful:
    - On a year-long project, every programmer will take two weeks of vacation
    - Buffers can be used to account for this known delay

Building the project schedule

- Don't Abuse Buffers
  - Buffers are often abused
    - The idea that overruns are expected means that there is an implicit assumption that the estimate is incorrect.
    - Buffers should not be used to add time to compensate for an inaccurate estimate.
- Example

"A programmer may find that she underestimated her programming task because the technical solution was more difficult than she originally anticipated. If there is a buffer, she may feel comfortable not reporting that mistake because she assumes that it will be absorbed. Unfortunately, if the designer and tester both made similar estimation mistakes and also failed to report them, all three of them are now counting on using the same buffer: there will be a large overrun. Since nobody reported any individual overruns, the project manager will not know about the problem until after the project has already slipped; this will limit his options much more than if he had known about the estimation problems as soon as they were discovered."

Project metrics

- Track the performance of the project
- It is important that the project team who will do the work should all agree that it represents a realistic plan for completing the project.
- The *baseline* is the version of the schedule that has been approved
  - The schedule will change based on the actual work done by the project team.
  - When the deadline of the revised schedule is later than that of the baseline, the project has *slipped*.

Project metrics cont.

- Therefore, there are two versions of the schedule.
- One is a fixed baseline version that is kept as a reference, and the other is an actual version of the schedule that is updated to reflect what actually happened over the course of the project.
- The baseline schedule never changes over the course of the project.
- Every time a task is delayed or changed, the actual schedule should be updated to reflect that.
- Each schedule change should be stored as a separate revision, which shows a snapshot of what the schedule looked like at any time in its history.
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Project metrics cont.

- A project not completed on due date does not tell the whole story.
- Schedule might slip because:
  - Team is waiting for a single person to complete a task
  - There is a general tendency to underestimate the effort required to perform all tasks
- Important for PM to track down source of each slip to improve estimation in future
- To understand nature of slop variance is calculated
- Variance is the difference between the estimated effort in the baseline and the actual effort performed by the team.

Project metrics cont.

- Variance is now the core of a project management system called earned value management, which tracks the project by considering effort "earned" against a budget only after it has actually been performed.

- For software projects, the variance is a measurement in person-hours (or person-days, person-years, etc.) that shows the difference between the effort planned to date on the baseline and the effort completed on the actual schedule.

Earned Value Analysis

- Earned Value Analysis or EVA is a quantitative technique for assessing the progress of a project.
- The earned value system provides a common value scale for every software task, regardless of the type of work being performed.
- The total hours to do the whole project are estimated, and every task is given an earned value based on the estimated percentage of the total.

Earned Value Analysis some terms

- Resource: The people, equipment, and material that are used to complete tasks in a project
- Fixed Costs: A set cost for a task that remains constant regardless of the task duration or the work performed by a resource
- BCWS (Budgeted Cost of Work Schedule): The earned value field that shows how much of the budget should have been spent, in view of the baseline cost of the task, assignment, or resource. BCWS is calculated as the cumulative time-phased baseline costs up to the status date or today's date

Earned Value Analysis some terms cont.

- Base line cost: The original project, resource, and assignment cost as shown in the baseline plan. The baseline cost is a snapshot of the cost at the time when the baseline plan was saved
- Status date: Date that you set [rather than the current date] for reporting the time, cost, or performance condition of a project

Earned Value Analysis cont.

At the root of earned value analysis are three fundamental values calculated for each task

1. The budgeted cost of tasks as scheduled in the project plan, based on the costs of resources assigned to those tasks, plus any fixed costs associated with the tasks .

- It is abbreviated as BCWS.
- BCWS is the baseline cost up to the status date you choose.
  - For example, the total planned budget for a 4-day task is Rs. 10,000 and it starts on a Monday. If the status date is set to the following Wednesday, the BCWS is Rs. 7,500.

Earned Value Analysis cont.

2. The actual cost required to complete all or some portion of the tasks, up to the status date.

- This is the actual cost of work performed (ACWP).
- For example, if the 4-day task actually incurs a total cost of 3500 during each of the first 2 days, the ACWP for this period is 7000 (but the BCWS is still 7500).

Earned Value Analysis cont.

3. The value of the work performed by the status date, measured in currency.

- This is literally the value earned by the work performed and is called the budgeted cost of work performed (BCWP).
- For example, if after 2 days 60% percent of the work on a task has been completed, you might expect to have spent 60 percent of the total task budget, or 6000.

Earned Value Analysis cont.

- Earned value analysis is always specific to a status date you choose.
- You may select the current date, a date in the past, or a date in the future.
- Most of the time, you'll set the status date to the date you last updated project progress. For **example**, if the current day is Tuesday, 12th of Sep, but the project was last updated with progress on Friday, 8th of Sep, you'd set the status date to Friday, 8th of September.

Earned Value Analysis Example

- Let's say a task has a budgeted cost (BCWS) of 10,000, and by the status date (set into the half of the task schedule) it is 40 percent complete. The earned value (BCWP) is 4000, but the scheduled value (BCWS) at the status date is 5000. This tells you that the task is behind schedule—less value has been earned than was planned.

Earned Value Analysis Example cont.

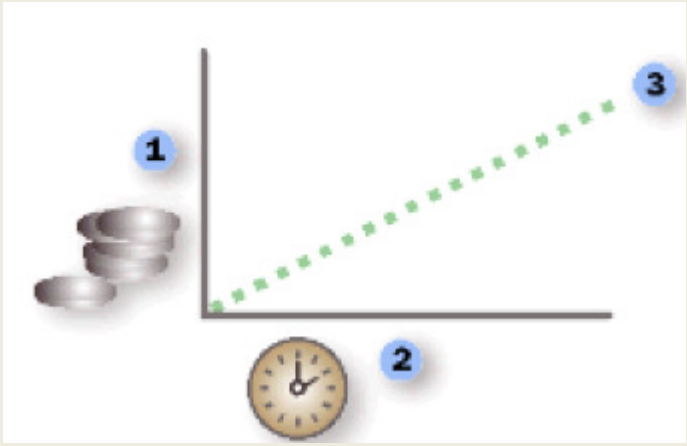
Let's also say that the task's actual cost (ACWP) at the status date is 6000, perhaps because a more expensive resource was assigned to the task. This tells you that the task is also over budget—more cost has been incurred than was planned.

You can see how powerful such an analysis can be. The earlier in a project's life cycle you identify such discrepancies between ACWP, BCWP and BCWS, the sooner you can take steps to remedy the problem.

Earned Value Analysis cont.

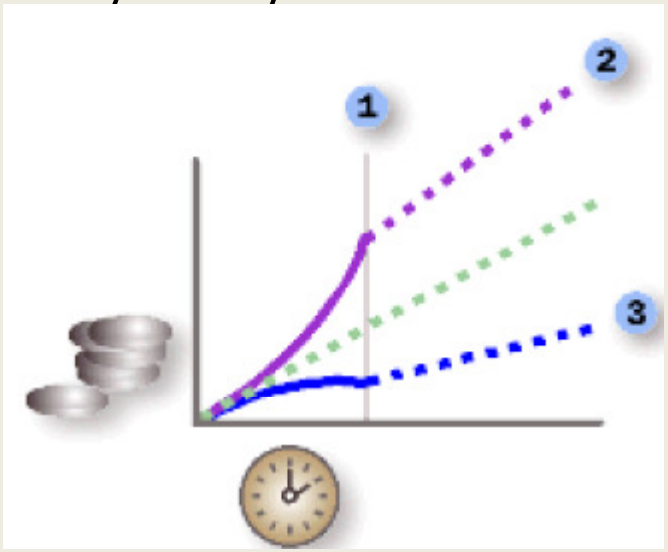
One common way of visualizing the key values of earned value analysis is to use a chart. Start with a simple chart showing a steady accumulation of cost over the lifetime of a project:

1. The vertical y-axis shows the projected cumulative cost for a project.
2. The horizontal x-axis shows time.
3. The planned budget for this project shows a steady expenditure over the lifetime of the project. This line represents the cumulative baseline cost.



Earned Value Analysis cont.

- After work on the project has begun, a chart of the key values of earned value analysis may look like this:



1. The status date determines the values Project calculates.
2. The actual cost (ACWP) of this project has exceeded the budgeted cost.
3. The earned value (BCWP) reflects the true value of the work performed. In this case, the value of the work performed is less than the amount spent to perform that work.

Earned Value Analysis cont.

- Earned value can also be used by generating the cost performance index (CPI) for the project

  CPI = BCWS / ACWP * 100 (percentage)

  100% means right cost, right budget

  < 100% means work cost less than planned

  > 100% means estimate was not adequate for the work involved

- CPI can be used either to compare projects or phases within a project